

SMART INDIA HACKATHON 2026

IDEA SUBMISSION & PRESENTATION DECK

PROJECT TITLE

WEIGHSURE METROLOGY ENGINE

Automated Verification & Digital Certification Platform for NAWI (OIML R-76 & Legal Metrology)

- **Problem Statement ID:** PS-35
- **Problem Statement Title:** Automated Verification & Digital Certification Platform for Non-Automatic Weighing Instruments (NAWI)
- **Theme:** Smart Automation / Governance & Consumer Protection
- **PS Category:** Software (Web & Field-Mobile Verification Platform)
- **Team ID:** [Registered SIH Team ID]
- **Team Name:** Team MetronIQ (Registered on portal)

PROBLEM STATEMENT & WEIGHSURE SOLUTION

The Problem Context:

- Manual Math Latency: Inspectors spend 35-45 mins manually computing multi-tier MPE error limits ($\pm 0.5e$, $\pm 1.0e$, $\pm 1.5e$) on paper registers.
- Statutory Misclassification: High rejection/approval errors due to confusing Initial Approval (1.0x MPE) with In-Service Stamping (2.0x MPE).
- Paper Fraud: Physical stamp certificates lack digital traceability, enabling counterfeit stamping and consumer exploitation in trade.

The WeighSure Solution:

- Instant OIML R-76 Automation: Ingests scale parameters (Class I-IV, Max, e, d) and executes multi-tier MPE checks in <100ms.
- Comprehensive 3-Test Suite: Evaluates Weighing Accuracy ($E = I - L$), Eccentricity (1/3 Max off-center), and Repeatability (10 cycles).
- Rule 14 Statutory Switch: One-click legal protocol toggle dynamically recalibrates between Initial Approval (Form VI) and In-Service Re-Stamping (Form VII).

KEY USPs & VERIFICATION PIPELINE

Key USPs (Why WeighSure Beats Competitor Solutions):

- Zero-Hallucination Math Core: Pure deterministic TypeScript/Python engine with IEEE-754 precision; zero generative AI math risk.
- Statutory Rule 14 Aware: Automates statutory doubling of Maximum Permissible Errors for routine field inspections under legal mandates.
- Quarantined AI Diagnostics: Gemini 2.5 Flash strictly diagnoses physical load cell tilt, corner binding, and fatigue without tampering calculations.
- Cryptographic Seal: Verifiable SHA-256 digital certificate hash & QR code prevent fake stamping certifications.

Step-by-Step Verification Pipeline:

1. Ingest Scale Specs (Accuracy Class I-IV, Max, e, d, Resolution n)
2. Environmental Baseline Audit (Temp, Humidity, Pressure Check)
3. Record Test Readings (Weighing, Eccentricity, Repeatability)
4. Deterministic Engine (Table 6 Envelope Calculation in <100ms)
5. AI Diagnostic Scan (Detect Load Cell Tilt & Corner Binding)
6. Digital Certificate (Schedule-VIII Form VI/VII with QR Code)

5-LAYER SYSTEM ARCHITECTURE

1. Presentation & Ingestion Layer

React 18 + Vite | Tailwind CSS | Lucide Icons
Guided 3-Step Audit Wizard with 44px field touch targets

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2. Deterministic Metrology Engine

Custom TypeScript & Python Metrology Core
Resolution: $n = \text{Max}/e$ | OIML Table 6 Tiers (500e, 2000e)
Formulas: $E = I - L$ | $E_c = E - E_o$ | Multiplier: $1.0x / 2.0x$ MPE

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3. Hardware Fault Diagnostic Layer

Google Gemini 2.5 Flash API
Analyzes Eccentricity Deltas & Strain-Gauge Creep Patterns
Provides actionable maintenance guidance to scale technicians

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4. Backend API & Persistence Layer

Python (FastAPI / Flask) High-Performance REST Services
SQLite Embedded Database (Zero-Config, Offline Relational)

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5. Statutory Verification & Audit Layer

Schedule-VIII Form VI & VII PDF Report Generator
SHA-256 Cryptographic Hash & Tamper-Proof Audit Trail

TECHNOLOGIES USED & RATIONALE

- React 18 + TypeScript** [Frontend Presentation Layer]
Enforces strict compile-time types on mass units (g, kg, mg) and divisions (e, d), preventing fatal floating-point precision errors during audits.
- Python (FastAPI / Flask)** [High-Performance API Backend]
High execution speed, native asynchronous handling, scientific math precision, and clean integration with metrology test pipelines.
- SQLite Database** [Embedded Local & Cloud Storage]
Zero-configuration, lightweight, serverless relational database; operates offline on field tablets with instant syncing to central state databases.
- Custom OIML Metrology Engine** [Statutory Math Module]
100% deterministic calculation engine without third-party dependencies; fully conforms to OIML R-76-1 Clause 3.5.1 with zero latency.
- Tailwind CSS** [Mobile-First Field Design System]
High-contrast UI designed for outdoor readability under harsh sunlight in rural APMC mandis and industrial weighbridges.
- Google Gemini 2.5 Flash** [Hardware Diagnostic Assistant]
Fast contextual telemetry reasoning that pinpoints mechanical faults (corner binding, load cell fatigue) when a scale fails verification.

FEASIBILITY ANALYSIS & RISK MITIGATION

Operational Feasibility Analysis:

- **Technical Feasibility:** Runs natively in any standard web browser (smartphone, tablet, laptop) without requiring native app installations.
- **Operational Feasibility:** Directly matches the statutory checklist and workflow used by Legal Metrology Officers (ILM) and RRSL laboratories.
- **Low Hardware Overhead:** Python backend with SQLite embedded database creates an ultra-lightweight footprint (<50MB RAM), operating on budget devices.

Key Challenges & Engineering Mitigations:

- **Challenge: Zero Connectivity in Rural Mandis**

Mitigation: Progressive Web App (PWA) with local SQLite/IndexedDB caching. Field officers complete audits offline; data auto-syncs when online.

- **Challenge: Field Officer Tech Reluctance**

Mitigation: Guided 3-step wizard with automatic boundary constraints that prevent invalid inputs (such as $d > e$ or negative loads) with zero learning curve.

- **Challenge: Binary Floating-Point Errors**

Mitigation: Python decimal module and custom rounding utilities fix calculations to 6 decimal places before evaluating against statutory scale divisions.

4 CORE VIABILITY PILLARS

1. Statutory & Regulatory Viability

Fully compliant with The Legal Metrology Act 2009 and Legal Metrology (General) Rules 2011. Generates authentic Schedule-VIII Form VI & VII verification certificates.

2. Economic Viability

Eliminates paper registers, physical log filing, and transit delays. Saves up to 80% officer hours per inspection, increasing daily scale audit capacity by 400%.

3. National Scalability Viability

Engineered for seamless API plug-and-play integration with the Department of Consumer Affairs national e-Maap portal and state enforcement databases.

4. Global Standard Viability

Because it strictly adheres to international OIML R-76 standards, the verification engine is immediately exportable across BIMSTEC and global metrology bodies.

MEASURABLE QUANTITATIVE IMPACT

85% Time Savings

Reduces total verification and calculation time from ~40 minutes down to under 5 minutes per scale.

0% Math Calculation Mistakes

Completely eliminates human computational errors in evaluating multi-tier MPE step boundaries.

100% Anti-Tamper Traceability

SHA-256 digital seals and GPS/time-stamped inspection records prevent counterfeit scale certificates.

4x Field Audit Throughput

Empowers each field officer to inspect up to 40 instruments daily, up from a previous average of 10.

Zero Paper Waste

Completely replaces bulky handwritten legal registers with searchable, cloud-backed audit records.

MULTI-STAKEHOLDER BENEFITS MATRIX

For Government & Legal Metrology Departments

- Real-time district-level dashboards tracking verification status, expired commercial seals, and statutory fee collections.
- Complete digital paper trail supporting the national Digital India mission.

For Consumers, Citizens & Farmers

- Guarantees fair weight in APMC mandis, fair-price ration shops (PDS), and neighborhood retail grocery stores.
- Public accountability: Anyone can scan the QR code on a scale sticker to verify its calibration status and expiration date.

For Scale Manufacturers, Dealers & Repairers

- Provides clear graphical failure reports showing the exact load point and corner where a scale failed, making recalibration quick and objective.
- Eliminates dispute delays through clear, standardized pass/fail metrics.

Statutory International Standards

OIML R-76-1 (Edition 2006 E):
Non-automatic weighing instruments - Part 1: Metrological and technical requirements - Tests (BIML, Paris).

OIML R-76-2 (Edition 2007 E):
Non-automatic weighing instruments - Part 2: Standard test report format.

Authenticated Metrology Datasets

CSIR - National Physical Laboratory (NPL, India):
Reference calibration data for Class E2, F1, M1 standard weights and transducer drift curves.

NIST Handbook 44 (2024 Edition):
Specifications, Tolerances, and Other Technical Requirements for Weighing Devices.

National Statutory Acts & Rules

The Legal Metrology Act, 2009 (Act No. 1 of 2010):
Sections 15, 24, and 30 governing inspection powers, mandatory verification, and penalties.

Legal Metrology (General) Rules, 2011:
Eighth Schedule (Form VI & Form VII) and Rule 14 (In-Service Double Tolerances).

Peer-Reviewed Academic Literature

Forbes, A. B. et al. (2021):
"Data analysis and uncertainty evaluation for legal metrology and weighing instruments", Metrologia (IOP Publishing), Vol. 58, No. 3.

Schwenke, H. et al. (2018):
"Digital transformation in legal metrology", PTB-Mitteilungen.

Measurement Uncertainty Guidelines

BIPM JCGM 100:2008 (GUM):
Guide to the expression of uncertainty in measurement, Joint Committee for Guides in Metrology.

IEEE Transactions on Instrumentation & Measurement:
Digital filtering and non-linear hysteresis compensation models for load cell sensors.

Regulatory & Software Guidelines

WELMEC Guide 7.2 (Software Guide):
Measuring Instruments Directive 2014/32/EU guidelines for legal metrology software security.

Department of Consumer Affairs (e-Maap Portal):
National guidelines for verification workflows and electronic weighing scale calibration.